Econ 602 Notes

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Noncooperative Bargaining

Model "Divide a dollar"

- Two players simultaneously submit demands $d_i \in [0, 1]$
- Continuum of Nash equilibria–this will lead us to look at SPE instead.

Def "Ultimatum Game"

- "sequential" but only one round (P1 makes TIOLI offer to P2 who accepts or rejects)
- **Prop:** has a **unique** SPE in which P1 offers (1,0) and P2 accepts any offer.
- multiple NE in the subgame following the offer, but P1 only has a BR if P2 accepts (1,0).

Model "Finite Alternating-Offer Bargaining"

Regularities:

1. Immediate agreement, efficient outcome
2. Proposer's offer makes responder indifferent
3. Unique SPE
4. Division of surplus depends:
   1. $x_1^*(T, \delta) = \sum_{t=1}^{T} (-\delta)^{t-1} = \frac{1-(-\delta)^T}{1+\delta}$
   2. $\delta \to 1 \Rightarrow$ last mover takes all
   3. $T \to \infty \Rightarrow X^* \to \frac{1}{1+\delta}$
   4. $T \to \infty$ then $\delta \to 1 \Rightarrow$ equal sharing

   Every game in this class has a unique subgame perfect equilibrium, and in it there is immediate agreement in the first round.

   The proposer in round T offers to take everything, and the responder accepts every offer

Thm "infinite alternating-offer bargaining game outcome"

The infinite horizon alternating-offer bargaining game with discount factor $\delta < 1$ has a unique subgame perfect equilibrium, in which Player 1 offers the payoff vector $(\frac{1}{1+\delta}, \frac{\delta}{1+\delta})$ in the first round and Player 2 accepts.

Finitely Repeated Games

Model "Repeated Games"

- Stage Game $\langle N, A, \pi \rangle$ (Normal form game)
- Players $N = \{1, ..., N\}$
- Action profiles $A = A_1 \times \cdots A_n$
- Stage payoffs $\pi : A \to \mathbb{R}^n$ (uniformly bounded) (i.e., $\pi_i(a)$ is i's payoff from action profile a)

- Periods $t = 1, ..., T$, where $T < \infty$
- Discount factor $\delta > 0 (\delta < 1$ if $T = \infty)$

"Histories and Strategies"

A strategy (in a repeated game) maps from the history of past play into an action for the current period.

- History $h^t \equiv (a^1, ..., a^{t-1})$, $h^1 \equiv \varnothing$
- Possible t-histories $\mathcal{H} \equiv A^{t-1}$
- Possible histories $\mathcal{H} \equiv \cup_{t=1}^{T} \mathcal{H}^t$
- Correlated behavior strategy profile $\sigma : \mathcal{H} \to \Delta A$
- A (pure) strategy is a function $s_i : \mathcal{H} \to A_i$
- a (behavioral) mixed strategy is defined in terms of strategy profiles to allow for correlation. So $\sigma : \mathcal{H} \to \Delta A$.

Def "Utility"

$$\text{Total Utility} = \tilde{u}_i(\sigma) \equiv \sum_{t=1}^{T} \delta^{t-1} \sum_{h^t \in \mathcal{H}^t} \pi_i(\sigma(h^t)) Pr(h^t|\sigma)$$

$$\text{Average Utility} = u_i(\sigma) \equiv \frac{1-\delta}{1-\delta^T} \sum_{t=1}^{T} \delta^{t-1} \sum_{h^t \in \mathcal{H}^t} \pi_i(\sigma(h^t)) Pr(h^t|\sigma)$$